

Multiple Regression

ECON20222 - Lecture 5

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Multiple Regression $K = 3$

Multiple (or multivariate) regression is when we have more than one RHS variable (or covariate or regressor). Consider:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u. \quad (1)$$

Assume we have a sample of data:

$$\{(y_i, x_{1i}, x_{2i}) : i = 1, \dots, n\}.$$

There are $K = 3$ parameters to estimate. We need three exogeneity assumptions (moment conditions):

$$E(u) = 0 \quad (\text{A1}) \quad (2)$$

$$\text{Cov}(u, x_1) = 0 \quad (\text{A2}) \quad (3)$$

$$\text{Cov}(u, x_2) = 0 \quad (\text{A3}) \quad (4)$$

Multiple Regression $K \geq 3$

The model extends to K covariates:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + u. \quad (5)$$

When multiple regression fails

Multiple regression fails when:

- 1 Any regressor has no variation:

$$\text{Var}(\hat{x}_{ki}) = 0.$$

- 2 There exists an exact linear combination between the x_{ki} s:

$$\lambda_1 x_{1i} + \lambda_2 x_{2i} = \lambda_0.$$

This is known as **perfect multicollinearity**.

Examples:

$$x_{1i} = -\lambda_2 x_{2i} \quad (\lambda_1 = 1, \lambda_0 = 0) \tag{6}$$

$$x_{1i} + x_{2i} = 1 \quad (\lambda_1 = \lambda_2 = \lambda_0 = 1) \tag{7}$$

Firm-size and wages

- Larger firms pay higher wages than smaller firms.
- In 20222 USoc `extract.dta`, there is a variable which records firm-size as one of $K = 9$ bands or bins or buckets::

$[1, 2], [3, 9], [10, 24], [25, 49], [50, 99], [100, 199],$
 $[200, 499], [500, 999], [1000, \infty)$

- `fsize` is actually an integer-valued ordered **multi-category variable** that records the mid-point of each bin:

1, 6, 17, 37, 75, 150, 350, 750, 1500.

Firm-size and wages

- The Key summary statistics

```
## =====
## Statistic   N      Mean   St. Dev.   Min Pctl(25) Pctl(75)   Max
## -----
## lnhrpay 58,789    2.284    0.631   -7.816    1.890    2.676   8.868
## fsize   58,789   303.522  484.630    1        17    350   1,500
## -----

## # A tibble: 9 x 4
##   fsize mean_lnhrpay sd_lnhrpay   n
##   <dbl>      <dbl>      <dbl> <int>
## 1     1         2.07         0.871  2235
## 2     6         2.04         0.590  7695
## 3    17         2.14         0.597  9575
## 4    37         2.21         0.612  9048
## 5    75         2.31         0.587  6746
## 6   150         2.37         0.600  5796
## 7   350         2.39         0.551  6782
## 8   750         2.48         0.568  3759
## 9  1500         2.61         0.615  7153
```

Firm-size and wages: Model

- There are two strategies: (a) non-parametric (b) parametric.
- For (a), we “create” $K = 9$ dummy variables:

$$x_{1i} = \begin{cases} 1 & \text{if } 1 \leq f_i^* \leq 2 \\ 0 & \text{else} \end{cases} \quad \text{or} \quad x_{1i} = \mathbf{1}(1 \leq f_i^* \leq 2)$$

$$x_{2i} = \begin{cases} 1 & \text{if } 3 \leq f_i^* \leq 9 \\ 0 & \text{else} \end{cases} \quad \text{or} \quad x_{2i} = \mathbf{1}(3 \leq f_i^* \leq 9)$$

$\vdots$

$$x_{9i} = \begin{cases} 1 & \text{if } f_i^* \geq 1000 \\ 0 & \text{else} \end{cases} \quad \text{or} \quad x_{9i} = \mathbf{1}(f_i^* \geq 1000)$$

- The x_k s are `fsize_f1`, `fsize_f6`, ..., `fsize_f1500` in Ralf's pdf.
- f_k is `mfsize9` (recoded to `fsize`).

Visualising the data

i	$\log y_i$	1	x_{1i}	x_{2i}	$\cdots$	x_{9i}	f_k	f_i^*
1	5.42	1	1	0	$\cdots$	0	1	1
2	5.39	1	1	0	$\cdots$	0	1	2
3	5.62	1	1	0	$\cdots$	0	1	2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$		
n_1	4.36	1	1	0	$\cdots$	0	1	1
$n_1 + 1$	4.29	1	0	1	$\cdots$	0	6	3
$n_1 + 2$	4.86	1	0	1	$\cdots$	0	6	7
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$		
$n_1 + n_2$	4.86	1	0	1	$\cdots$	0	6	8
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$		
$\vdots$	3.42	1	0	0	$\cdots$	1	1500	1324
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$		
n	3.33	1	0	0	$\cdots$	1	1500	7022

Dummy Variable Trap

- The regression we wish run is written:

$$\log y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_9 x_{9i} + u_i.$$

- However, the computer cannot compute OLS estimates because

$$x_{1i} + x_{2i} + \cdots + x_{9i} = 1 \quad \text{for every individual } i.$$

This example of **perfect multi-collinearity** is also known as the **dummy variable trap**.

- We therefore drop a dummy, and actually choose the largest firm-size category:

$$\log y = \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_8 x_8 + \beta_0 + u. \quad (8)$$

Empirical Results

```
## =====
##                               Dependent variable:
##                               -----
##                               lnhrpay
## -----
##  fsize_f1                    -0.548***
##                               (0.015)
##  fsize_f6                    -0.579***
##                               (0.010)
##  fsize_f17                   -0.472***
##                               (0.009)
##  fsize_f37                   -0.409***
##                               (0.010)
##  fsize_f75                   -0.310***
##                               (0.010)
##  fsize_f150                  -0.242***
##                               (0.011)
##  fsize_f350                  -0.229***
##                               (0.010)
##  fsize_f750                  -0.138***
##                               (0.012)
##  Constant                    2.615***
##                               (0.007)
##
## -----
##  Observations                58,789
##  R2                          0.080
##  Adjusted R2                 0.080
##  Residual Std. Error         0.606 (df = 58780)
##  F Statistic                  636.491*** (df = 8; 58780)
##  =====
##  Note: *p<0.1; **p<0.05; ***p<0.01
```

Interpreting the regression

- Repeating the $K = 2$ case in Lecture 3, the β s are interpreted as:

$$\beta_1 \equiv \mathbb{E}(\log y \mid x_1 = 1) - \mathbb{E}(\log y \mid x_9 = 1)$$

$$\vdots$$

$$\beta_8 \equiv \mathbb{E}(\log y \mid x_8 = 1) - \mathbb{E}(\log y \mid x_9 = 1)$$

$$\beta_0 \equiv \mathbb{E}(\log y \mid x_9 = 1)$$

- ▶ The β s can also be written:

$$\beta_1 \equiv \mathbb{E}(\log y \mid 1 \leq f^* \leq 2) - \mathbb{E}(\log y \mid f^* \geq 1000)$$

$$\vdots$$

$$\beta_8 \equiv \mathbb{E}(\log y \mid 500 \leq f^* \leq 999) - \mathbb{E}(\log y \mid f^* \geq 1000)$$

$$\beta_0 \equiv \mathbb{E}(\log y \mid f^* \geq 1000)$$

Interpreting the regression

- In words, β_0 is average log real hourly wage for the omitted or base category (working for a firm employing 1000+ workers).
- All the other β_k s are interpreted as the population raw differential in log real hourly wages between the firm-size category that β_k “belongs to” and the largest firm-size category.
- These are estimated by the corresponding raw differentials and are measured in log-points.

Estimating the regression

- The results are in Ralf's `Regression_Multiple.html`.
- See the second regression for the estimation of (8).
- The parameters are interpreted as above. The estimate on the constant is 2.615 (0.007) and is average log-pay for those working in the largest firm-size category (1000+).
- The estimate on `fsize_f1` is -0.548 and is the raw differential in log real hourly wage between those in the smallest firm-size category (1–2) and those in the largest firm-size category (1000+). It is large, and converts to a differential of about -42.2% .
- In both cases, exactly the same estimates can be calculated from the `tibble`.

Do we really have 59k data points?

- Just as there were 2 population means when we have a gender dummy, here we have:

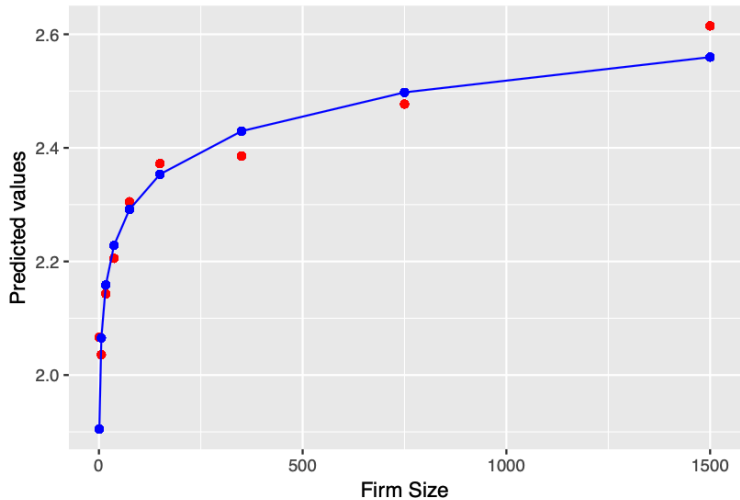
$$\mathbb{E}(\log y \mid k = j), \quad j = 1, \dots, 9.$$

- This means there are just 9 fitted values, depending on which firm-size category the individual belongs to.
- But there are 58,789 individual-years (datapoints).
- So one can think of regression as “collapsing” the data to just 9 data-points:

$$\left\{ \left(\overline{\log y_k}, f_k \right) : k = 1, \dots, 9 \right\}.$$

- See the earlier `tibble`. All the regression does is replicate this information.
- So what is the benefit of having micro-econometric data? Each cell-mean $\overline{\log y_k}$ is estimated from n_k obs. Without knowing n_k , inference is not possible.

Predicted Regression Model



Modelling the 9 datapoints

- In the next 4 slides, we model these 9 datapoints.
- These are the red blobs. They seem to follow some non-linear relationship.
- What is the blue line? It is the second **parametric** strategy we could adopt for modelling this relationship. Here we choose a log-log model:

$$\log y = \delta_0 + \delta_1 \log f + \text{error}.$$

Empirical results

```
## =====
##                               Dependent variable:
##                               -----
##                               lnhrpay
## -----
## log(fsize)                    0.090***
##                               (0.001)
##
## Constant                      1.905***
##                               (0.006)
##
## -----
## Observations                  58,789
## R2                            0.075
## Adjusted R2                   0.075
## Residual Std. Error          0.607 (df = 58787)
## F Statistic                   4,763.519*** (df = 1; 58787)
## =====
##                               Note: *p<0.1; **p<0.05; ***p<0.01
##                               Robust standard errors in parenthesis
```

Alternative parametric models

- One can see that log–log relationship is a reasonable fit. The estimate of $\hat{\delta}_1$ is 0.0896 (0.0014). It is an elasticity (unit-free).
- Other **parametric** models could be fitted, eg a quadratic:

$$\log y = \delta_0 + \delta_1 f + \delta_2 f^2 + \text{error}.$$

The relationship between log-pay and age is often modelled like this.

Conclusion

- Multiple regression is the basic tool in econometrics.
- We have illustrated its use in just one example.
- Many more examples will appear later.
- It seems to be the case that large firms pay more than smaller firms. But only the raw data have been used.
- So we cannot interpret this relationship as causal. We could and should add more covariates, like age and education, but . . .
- OLS cannot deal the fact that individuals choose where to work, and know that larger firms pay more than smaller ones, *ceteris paribus*. Similarly, good firms will choose to hire workers with high innate ability (unobserved) and pay them more.

Controlling for observables

- The basic idea is that we add as many relevant variables as we can to a model, to mitigate/minimise some of the effect of omitted variables bias (OVB). Recall Lecture 4.
- But there is often something fundamentally unobserved.
- In

$$\begin{aligned}y &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_K x_K + u \\ &= \beta_0 + \beta_1 x_1 + \mathbf{x}'\mathbf{b} + u\end{aligned}$$

β_1 is the effect of interest. All the other variables are **controls**.

- Notation:

$$\mathbf{x}' = (x_2, \dots, x_K) \quad \mathbf{b}' = (\beta_2, \dots, \beta_K).$$

- Often x_1 is a dummy d .
- A concrete example is at the end of this topic.

Long and short regressions

- When estimating

$$y = \beta_0 + \beta_1 d + \mathbf{x}'\mathbf{b} + u$$

the conditional differential is

$$\mathbb{E}(y \mid \mathbf{x}', d = 1) - \mathbb{E}(y \mid \mathbf{x}', d = 0).$$

- When estimating

$$y = \beta_0 + \beta_1 d + u$$

the unconditional differential is

$$\mathbb{E}(y \mid d = 1) - \mathbb{E}(y \mid d = 0).$$

- But do the OLS estimates differ from each other? Define $\hat{\beta}_1$ as the OLS estimate from the long regression and $\tilde{\beta}_1$ as the OLS estimate from the short regression.
- When the long contains one extra regressor we already know what happens (almost) ...

OVB again

- Compare the estimates from:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e \quad (\text{L})$$

$$y = \beta_0 + \beta_1 x_1 + u, \quad (\text{S})$$

with $\text{Cov}(e, x_1) = \text{Cov}(e, x_2) = \mathbb{E}(e) = 0$.

- When estimating (S), the OVB formula from Lecture 4 says:

$$\text{Bias}(\tilde{\beta}_1) \equiv \mathbb{E}(\tilde{\beta}_1) - \beta_1 = \beta_2 \frac{\text{Cov}(x_2, x_1)}{\text{Var}(x_1)} = \beta_2 \delta_1,$$

where δ_1 is the “slope” parameter:

$$x_2 = \delta_0 + \delta_1 x_1 + \text{error}.$$

- When estimating (L), the OVB formula says the bias is zero:

$$\text{Bias}(\hat{\beta}_1) \equiv \mathbb{E}(\hat{\beta}_1) - \beta_1 = 0.$$

OVV again



$$\mathbb{E}(\tilde{\beta}_1) - \mathbb{E}(\hat{\beta}_1) = \beta_2 \delta_1.$$

- It turns out that this holds exactly in the data:

$$\tilde{\beta}_1 - \hat{\beta}_1 = \hat{\beta}_2 \hat{\delta}_1.$$

- When the long regression is longer than 2 covariates

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_K x_K + e, \quad (\text{L}')$$

the formula does not hold exactly, but it still works in principle.

- This formula is very, very useful.

Lifetime earnings and a university degree

- Consider the model:

$$y = \beta_0 + \tau d + \beta_1 a + \beta_2 a^2 + \mathbf{x}'\mathbf{b} + u$$

where y is earnings, d is a graduate dummy, and a is age.

- If we assume all the RHS variables are exogenous:

$$\mathbb{E}(y \mid a, \mathbf{x}', d = 1) = \beta_0 + \tau + \beta_1 a + \beta_2 a^2 + \mathbf{x}'\mathbf{b}$$

$$\mathbb{E}(y \mid a, \mathbf{x}', d = 0) = \beta_0 + \beta_1 a + \beta_2 a^2 + \mathbf{x}'\mathbf{b}$$

and so

$$\tau = \mathbb{E}(y \mid a, \mathbf{x}', d = 1) - \mathbb{E}(y \mid a, \mathbf{x}', d = 0)$$

is the so-called **conditional differential** (or **conditional treatment effect** if appropriate).

- Conceptually different from the **unconditional differential**

$$\mathbb{E}(y \mid d = 1) - \mathbb{E}(y \mid d = 0).$$

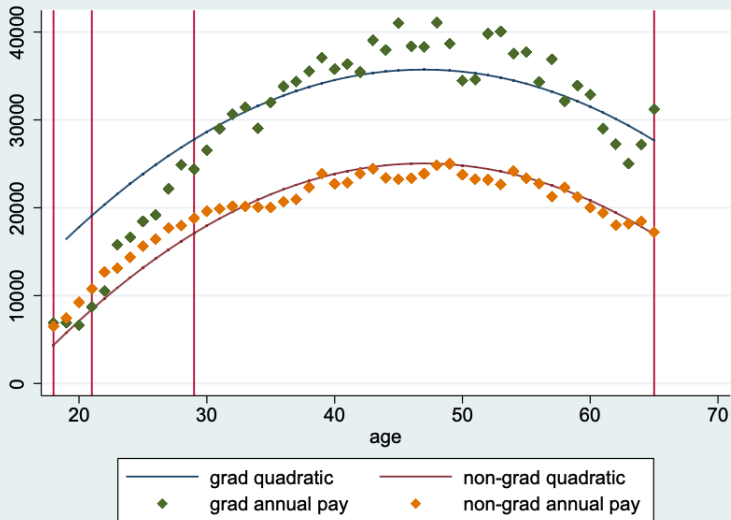
Two parallel CEFs

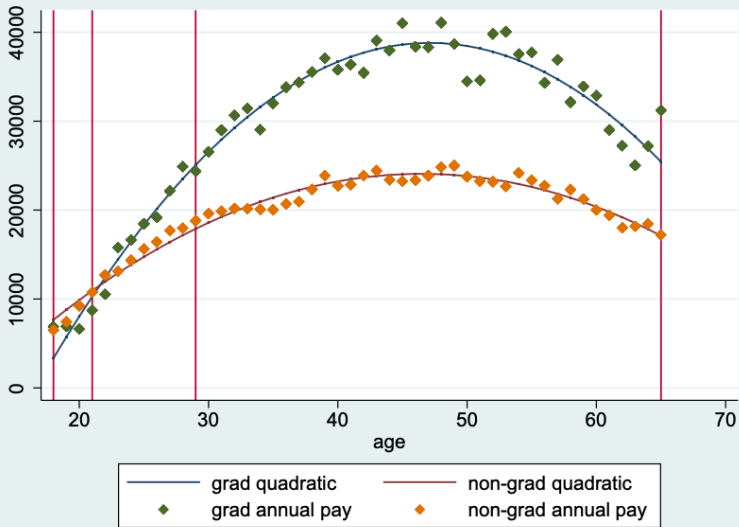
- Because everything comes from the same regression, everything nets out, especially age a .
- In other words, the 2 so-called **Conditional Expectation Functions** (CEFs) are parallel when plotted in the same picture.
- Plotting them can be pain because of the term $\mathbf{x}'\mathbf{b}$, after its been estimated:

$$\mathbf{x}'\hat{\mathbf{b}} = \hat{\beta}_2 \bar{x}_2 + \cdots + \hat{\beta}_K \bar{x}_K$$

But it only affects the constant.

- The next slide shows what happens in the BHPS/US.
- But is the slide after more realistic?





Two non-parallel CEFs

- How do we make the age-relationship vary by d ? Add 2 interaction terms:

$$y = \beta_0 + \beta_1 a + \beta_2 a^2 + \tau_0 d + \tau_1 da + \tau_2 da^2 + \mathbf{x}'\mathbf{b} + u.$$

- Rewriting the CEFs:

$$\mathbb{E}(y \mid a, \mathbf{x}', d = 1) = \beta_0 + \beta_1 a + \beta_2 a^2 + \tau_0 + \tau_1 a + \tau_2 a^2 + \mathbf{x}'\mathbf{b}$$

$$\mathbb{E}(y \mid a, \mathbf{x}', d = 0) = \beta_0 + \beta_1 a + \beta_2 a^2 + \mathbf{x}'\mathbf{b}$$

and so

$$\mathbb{E}(y \mid a, \mathbf{x}', d = 1) - \mathbb{E}(y \mid a, \mathbf{x}', d = 0) = \tau_0 + \tau_1 a + \tau_2 a^2.$$

- This can be computed at any age. The IFS do this at $a = 29$.
- For a Lifetime Earnings measure:

$$\int_{a=18}^{65} (\tau_0 + \tau_1 a + \tau_2 a^2) da = \left[\tau_0 a + \frac{\tau_1}{2} a^2 + \frac{\tau_2}{3} a^3 \right]_{18}^{65}.$$

Table 1: Male graduate and non-graduate annual pay and quadratic age

	Dependent variable: real annual pay				
	(1) raw	(2) parallel	(3) nongrad	(4) grad	(5) nonpar
graduate dummy (d)	11180 (258)	10679 (248)			-32387 (2664)
age (a)		2319 (48)	1883 (45)	3941 (138)	1883 (45)
age-squared ($a^2/100$)		-2470 (60)	-2026 (57)	-4182 (173)	-2026 (57)
grad dummy * age (da)					2059 (145)
grad dummy * age-sq ($da^2/100$)					-2156 (182)
constant	20267 (103)	-29368 (853)	-19665 (6785)	-54053 (2546)	-19665 (785)
Observations	25382	25382	17184	8198	25382
Adjusted R2	0.091	0.181	0.105	0.120	0.195

a. Robust standard errors in parentheses.

b. Data is the the BHPS/US extract discussed elsewhere.

c. Column (5) reports estimates of models on Slide 25.

d. Columns (3,4) are identical to Column (5), seen by switching the graduate dummy on-and-off.

Walker and Zhu

- "Differences by degree: Evidence of the net financial rates of return to undergraduate study for England and Wales"
- WZ extend the framework as follows. First write our non-parallel models as:

$$\log y = (\beta_0 + \tau_0) + (\beta_1 + \tau_1)a + (\beta_2 + \tau_2)a^2 + \mathbf{x}'\mathbf{b} + u \quad \text{if } d = 1$$

$$\log y = \beta_0 + \beta_1 a + \beta_2 a^2 + \mathbf{x}'\mathbf{b} + u \quad \text{if } d = 0$$

- Instead of 2 categories, we could have Q :

$$\log y_q = \beta_0^q + \beta_1^q a_q + \beta_2^q a_q^2 + \sum_j \lambda^{jq} D_q^j + \mathbf{x}'_q \mathbf{b}^q + u_q, \quad q = 1, \dots, Q$$

- Instead of looking at graduate ($d = 1$) and else ($d = 0$), WZ condition on those eligible to go to university and split these by q :
 - ▶ STEM, LEM, COMB, OSSAH; first class, upper second, below upper second, degree class missing.

Walker and Zhu

- This is WZ's Equation (2), except ours is a population model whereas there's is a sample model (data are indexed by i).
- The D^j 's are dummies for 'qualification' category/level: doctoral, master, PGCE, ..., first degree, 2+ A-levels.
- Hence the λ^{jq} 's are:

$$\lambda^{jq} = \mathbb{E}(\log y \mid \mathbf{x}', \text{qual} = j) - \mathbb{E}(\log y \mid \mathbf{x}', \text{qual} = 0) \quad \text{for } q = 1, \dots, Q$$

- WZ use the Labour Force Survey (LFS), whereas the above is from BHPS/US. Why?: much better info on degree.
- The LFS is a rolling panel (each i observed 5 quarters then exit). BHPS/US is 'proper' panel.
- WZ split by gender throughout (absolutely standard) whereas my results above are for males.

Walker and Zhu

- WZ model the real hourly wage rate, logged, whereas I model real annual pay, unlogged. WZ also worry about lifetime earnings, NPVs, IRRs.
- WZ write down their model in terms of years of labour market experience (the latent variable) but proxy it by age. This raises issues relating to measurement error biases. I model age from the outset.
- There are lots of tables of descriptives (xtabs), figures, and regressions. This is because their analysis is multi-dimensional, relating to:
 - ▶ educational attainment/level
 - ▶ degree classification
 - ▶ degree subject
 - ▶ gender
 - ▶ age/experience